

Muvvala Krishna Kiriti

 E40R |  Muvvala Kiriti |  muvvalakiriti@gmail.com |  +91 7013378027

EDUCATION

Vellore Institute of Technology, Bhopal B.Tech ECE with Specialization in AI and Cybernetics	<i>CGPA: 8.33/10</i> <i>2022 - 2026</i>
--	--

SKILLS

Programming & Data	Python, HTML, Java, SQL, NumPy, Pandas, Matplotlib, Scikit-learn, Pytorch
AI & ML & DL	DL:CNN RNN,OpenCV, Modular ML Pipelines, Data preprocessing
NLP & LLM	LangChain, Large Language Models, RAG, BERTopic, Semantic Search, Embeddings, Tokenization, Prompt Engineering, QGIS, GEE, ASF GRD
Containers & Cloud	AWS (Certificate link) , Azure, OCI , Docker

EXPERIENCE

Machine Learning Intern, NRSC/ISRO	Feb 2026 - Present
Built flood segmentation system on Sentinel-1 SAR imagery by fine-tuning SSL pretrained (MoCo,DINO) ResNet50/ViT encoders within a UNet framework, with gradual unfreezing, differential LRs, and combined Focal+Tversky loss for severe class imbalance. Used QGIS, Rasterio, GEE.	
AI Intern, PawsPalConnect	Feb 2026 - Present
- Built a real-time voice AI assistant with end-to-end streaming (FastAPI Websockets) pipeline integrating STT/TTS, Silero VAD, and LLM for structured clinical triage reasoning with barge-in interruption. - Promoted to AI Team Lead based on contributions to system design and deployment.	
AI Intern, Infosys Springboard	Sep 2025 - Dec 2025
Developed an AI-powered audio book generator using advanced NLP for text enrichment and open-source text-to-speech synthesis.	

PROJECTS

Diabetic Retinopathy Detection using CNNs & Explainable AI	<i>Sep 2025-Ongoing</i>
Python, TensorFlow, Keras, OpenCV, LIME/SHAP, Medical Imaging	
- Developed CNN-based classification system to detect 5 stages of DR from retinal fundus images. - Implemented LLM and XAI techniques (LIME) to provide model interpretability for clinical decision support. - Applied data preprocessing (cropping and resizing) and data augmentation strategies.	
Real-Time Facial Emotion Recognition using CNNs	<i>Dec 2024</i>
Python, OpenCV, TensorFlow, Keras	
- Designed and trained a CNN to classify facial expressions into 7 emotion categories. - Applied Data Augmentation techniques to improve model generalization, achieving 92% accuracy.	
RAG-based Efficient Customer Support with Semantic Analytics	<i>Oct 2025</i>
Python, LangChain, Hugging Face, Pinecone, Streamlit, API, Agile methodology	
- Engineered a customer support solution using RAG to process 1,000+ knowledge base documents. - Developed a real-time escalation dashboard with chat summaries, 10+ issue tags.	

HACKATHONS, AWARDS & RESEARCH

- Co-authored Springer publication *“Education AI”* (2025) [DOI](#) on AI-powered educational systems.
- Runner-ups – IIT KGP Empresario & ISB Neovate:**Built and presented our MVP, GTM.
- Winner, Blockchain Debate organized by Zeeve (2023)